

Coverage Report for Gate_Harness

Table of Contents

- 1. [Analysis Information](#)
- 2. [Aggregated Tests](#)
- 3. [Summary](#)
- 4. [Details](#)

Analysis Information

Coverage Data Information

Collected in version (R2022b)

Model Information

Model version 1.64
Author chiar
Last saved Wed Jun 05 09:36:15 2024

Harness information

Harness model(s) Gate_Harness
Harness model owner ElectronicGate

Simulation Optimization Options

Default parameter behavior inlined
Block reduction forced off
Conditional branch optimization on

Coverage Options

Analyzed model Gate_Harness/SYSTEM CORE
Logic block short circuiting off

Aggregated Tests

Run	Test Name	Date
Subsystem: "/SYSTEM CORE"		
U1.1	Run 1	05-Jun-2024 09:39:28

U1.2	Run 2	05-Jun-2024 09:39:42
U1.3	Run 3	05-Jun-2024 09:41:27
U1.4	Run 4	05-Jun-2024 09:41:32
U1.5	Run 5	05-Jun-2024 09:41:37
U1.6	Run 6	05-Jun-2024 09:41:42
U1.7	Run 7	05-Jun-2024 09:41:48
U1.8	Run 8	05-Jun-2024 09:41:52
U1.9	Run 9	05-Jun-2024 09:41:56
U1.10	Run 10	05-Jun-2024 09:42:00
U1.11	Run 11	05-Jun-2024 09:42:04
U1.12	Run 12	05-Jun-2024 09:42:08
U1.13	Run 13	05-Jun-2024 09:42:13
U1.14	Run 14	05-Jun-2024 09:42:17
U1.15	Run 15	05-Jun-2024 09:42:24
U1.16	Run 16	05-Jun-2024 09:42:29
U1.17	Run 17	05-Jun-2024 09:42:34
U1.18	Run 18	05-Jun-2024 09:42:40

Summary

Model Hierarchy/Complexity

		Decision		Execution
1.	SYSTEM CORE	70	89%	NA
2. . . .	SF: SYSTEM CORE	69	89%	NA
3.	SF: SYSTEM	69	89%	NA
4.	SF: B1	5	89%	NA
5.	SF: B2	5	89%	NA
6.	SF: B3	5	89%	NA
7.	SF: CLOSING TIME	5	100%	NA
8.	SF: GATE	44	88%	NA
9.	SF: ACTIVE	41	86%	NA
10.	SF: CLOSE	8	90%	NA
11.	SF: OPEN	8	90%	NA
12.	SF: WORKING TIME	5	100%	NA

Details

1. SubSystem block "[SYSTEM CORE](#)"

Child Systems: [SYSTEM CORE](#)

Metric	Coverage (this object)	Coverage (inc. descendants)
Cyclomatic Complexity	1	70
Decision	NA	89% (85/95) decision outcomes
Execution	NA	NA

2. Chart "[SYSTEM CORE](#)"

[Justify or Exclude](#)

Parent: [Gate_Harness/SYSTEM CORE](#)

Child Systems: [SYSTEM](#)

Metric	Coverage (this object)	Coverage (inc. descendants)
Cyclomatic Complexity	0	69
Decision	NA	89% (85/95) decision outcomes

3. State "[SYSTEM](#)"

[Justify or Exclude](#)

Parent: [Gate_Harness/SYSTEM CORE](#)

Child Systems: [B1](#), [B2](#), [B3](#), [CLOSING_TIME](#), [GATE](#), [WORKING_TIME](#)

Metric	Coverage (this object)	Coverage (inc. descendants)
Cyclomatic Complexity	0	69
Decision	NA	89% (85/95) decision outcomes

4. State "[B1](#)"

[Justify or Exclude](#)

Parent: [Gate_Harness/SYSTEM CORE.SYSTEM](#)

Metric	Coverage (this object)	Coverage (inc. descendants)
Cyclomatic Complexity	2	5
Decision	100% (3/3) decision outcomes	89% (8/9) decision outcomes

Decisions analyzed

Substate executed	100%
State "LONG_PRESSED"	24/7248 U1.1
State "PRESSED"	240/7248 U1.1
State "RELEASED"	6984/7248

Transition "[b1 == OFF]" from "LONG_PRESSED" to "RELEASED"

[Justify or Exclude](#)

Parent: [Gate_Harness/SYSTEM CORE.SYSTEM.B1](#)

Uncovered Links: 

Metric	Coverage
Cyclomatic Complexity	1
Decision	50% (1/2) decision outcomes

1 [b1 == OFF]

#1: [b1 == OFF]

Decisions analyzed

b1 == OFF	50%
false	0/24 
true	24/24 U1.1

Full Coverage

Model Object	Metric
Transition "[b1 == OFF]" from " <u>PRESSED</u> " to " <u>LONG_PRESSED</u> "	Decision
Transition "[b1==ON]" from " <u>RELEASED</u> " to " <u>PRESSED</u> "	Decision

5. State "B2"

[Justify or Exclude](#)

Parent: [Gate_Harness/SYSTEM CORE.SYSTEM](#)

Metric	Coverage (this object)	Coverage (inc. descendants)
Cyclomatic Complexity	2	5
Decision	100% (3/3) decision outcomes	89% (8/9) decision outcomes

Decisions analyzed

Substate executed	100%
State "LONG_PRESSED"	12/7248 U1.6

State "PRESSED"	60/7248 U1.6
State "RELEASED"	7176/7248 U1.1

Transition "[b2 == OFF]" from "[LONG_PRESSED](#)" to "[RELEASED](#)"

[Justify or Exclude](#)

Parent: [Gate_Harness/SYSTEM CORE.SYSTEM.B2](#)

Uncovered Links: 

Metric	Coverage
Cyclomatic Complexity	1
Decision	50% (1/2) decision outcomes

[1](#) [b2 == OFF]

[#1: \[b2 == OFF\]](#)

Decisions analyzed

b2 == OFF	50%
false	0/12 
true	12/12 U1.6

Full Coverage

Model Object	Metric
Transition "[b2==ON]" from " RELEASED " to " PRESSED "	Decision
Transition "[b2 == OFF]" from " PRESSED " to " LONG_PRESSED "	Decision

6. State "[B3](#)"

[Justify or Exclude](#)

Parent: [Gate_Harness/SYSTEM CORE.SYSTEM](#)

Metric	Coverage (this object)	Coverage (inc. descendants)
Cyclomatic Complexity	2	5
Decision	100% (3/3) decision outcomes	89% (8/9) decision outcomes

Decisions analyzed

Substate executed	100%
State "LONG_PRESSED"	12/7248 U1.5
State "PRESSED"	60/7248 U1.5
State "RELEASED"	7176/7248 U1.1

Transition "[\[b3 == OFF\]](#)" from "[LONG_PRESSED](#)" to "[RELEASED](#)"

[Justify or Exclude](#)

Parent: [Gate_Harness/SYSTEM CORE.SYSTEM.B3](#)

Uncovered Links: 

Metric

Coverage

Cyclomatic Complexity

1

Decision

50% (1/2) decision outcomes

[1](#) [**b3 == OFF**]

[#1: \[b3 == OFF\]](#)

Decisions analyzed

b3 == OFF	50%
false	0/12 
true	12/12 U1.5

Full Coverage

Model Object

Metric

Transition "[\[b3 == OFF\]](#)" from "[PRESSED](#)" to "[LONG_PRESSED](#)"

Decision

Transition "[\[b3==ON\]](#)" from "[RELEASED](#)" to "[PRESSED](#)"

Decision

7. State "[CLOSING_TIME](#)"

[Justify or Exclude](#)

Parent: [Gate_Harness/SYSTEM CORE.SYSTEM](#)

Metric	Coverage (this object)	Coverage (inc. descendants)
Cyclomatic Complexity	1	5
Decision	100% (2/2) decision outcomes	100% (6/6) decision outcomes

Decisions analyzed

Substate executed	100%
State "CLOSING_TIME_INC"	371/7248 U1.6
State "CLOSING_TIME_RESET"	6877/7248 U1.1

Full Coverage

Model Object

Transition "[\[b2_pressed & p2 == OFF\]](#)" from "[CLOSING_TIME_INC](#)" to "[CLOSING_TIME_INC](#)"

Metric

Decision

Transition "[\[b2_pressed & p2 == OFF\]](#)" from "[CLOSING_TIME_RESET](#)" to "[CLOSING_TIME_INC](#)"

Decision

8. State "[GATE](#)"

[Justify or Exclude](#)

Parent: [Gate_Harness/SYSTEM CORE.SYSTEM](#)

Child Systems: [ACTIVE](#)

Metric	Coverage (this object)	Coverage (inc. descendants)
Cyclomatic Complexity	1	44
Decision	100% (2/2) decision outcomes	88% (49/56) decision outcomes

Decisions analyzed

Substate executed	100%
State "ACTIVE"	6728/7248 U1.1
State "INACTIVE"	520/7248 U1.1

Full Coverage

Model Object

Transition "[\[start == ON\]](#)" from "[INACTIVE](#)" to "[ACTIVE](#)"

Metric

Decision

Transition "[start == OFF]" from "[ACTIVE](#)" to "[INACTIVE](#)"

Decision

9. State "[ACTIVE](#)"

[Justify or Exclude](#)

Parent: [Gate_Harness/SYSTEM CORE.SYSTEM.GATE](#)

Child Systems: [CLOSE](#), [OPEN](#)

Uncovered Links: 

Metric	Coverage (this object)	Coverage (inc. descendants)
Cyclomatic Complexity	10	41
Decision	58% (7/12) decision outcomes	86% (43/50) decision outcomes

Decisions analyzed

Substate executed	100%
State "CLOSE"	3310/6727 U1.1
State "ERROR_STATE"	602/6727 U1.6
State "IS_CLOSING"	469/6727 U1.1
State "IS_OPENING"	551/6727 U1.1
State "OPEN"	1594/6727 U1.1
State "STOP_SYSTEM"	201/6727 U1.6
Substate exited when parent exits	17%
State "CLOSE"	0/1 
State "ERROR_STATE"	0/1 
State "IS_CLOSING"	0/1 
State "IS_OPENING"	1/1 U1.4
State "OPEN"	0/1 
State "STOP_SYSTEM"	0/1 

Full Coverage

Model Object	Metric
Transition "[p2 == OFF]" from "ERROR_STATE" to "CLOSE"	Decision
Transition "[b1_pressed (p1 == OFF)]" from "IS_CLOSING" to "IS_OPENING"	Decision
Transition "[b1_pressed & (p1 == ON)]" from "IS_OPENING" to "IS_CLOSING"	Decision
Transition "[p2 == OFF]" from "IS_CLOSING" to "CLOSE"	Decision
Transition "after(working_time, sec)" from "IS_OPENING" to "OPEN"	Decision
Transition "[b1_pressed & p1 == ON] duration(p1 == ...)" from "OPEN" to "IS_CLOSING"	Decision
Transition "[b1_pressed & (p1 == ON)]" from "CLOSE" to "IS_OPENING"	Decision
Transition "[after(working_time, sec) & (p2 == ON)]" from "IS_CLOSING" to "STOP_SYSTEM"	Decision
Transition "after(WAITING_TIME, sec)" from "STOP_SYSTEM" to "ERROR_STATE"	Decision

10. State "CLOSE"

Justify or Exclude

Parent: [Gate_Harness/SYSTEM CORE.SYSTEM.GATE.ACTIVE](#)

Uncovered Links: 

Metric	Coverage (this object)	Coverage (inc. descendants)
Cyclomatic Complexity	2	8
Decision	75% (3/4) decision outcomes	90% (9/10) decision outcomes

Decisions analyzed

Substate executed	100%
State "GREEN_BLK_OFF"	2905/3300 U1.1
State "GREEN_BLK_ON"	395/3300 U1.14
Substate exited when parent exits	50%
State "GREEN_BLK_OFF"	10/10 U1.1
State "GREEN_BLK_ON"	0/10 

Full Coverage

Model Object	Metric
Transition "[after(GREENDUR, sec).](p1 == ON)]" from " GREEN_BLK_ON " to " GREEN_BLK_OFF "	Decision
Transition "[b1_pressed & (p1 == OFF)]" from " GREEN_BLK_ON " to " GREEN_BLK_ON "	Decision
Transition "[b1_pressed & (p1 == OFF)]" from " GREEN_BLK_OFF " to " GREEN_BLK_ON "	Decision

11. State "[OPEN](#)"

[Justify or Exclude](#)

Parent: [Gate_Harness/SYSTEM CORE.SYSTEM.GATE.ACTIVE](#)

Uncovered Links: 

Metric	Coverage (this object)	Coverage (inc. descendants)
Cyclomatic Complexity	2	8
Decision	75% (3/4) decision outcomes	90% (9/10) decision outcomes

Decisions analyzed

Substate executed	100%
State " GREEN_BLK_OFF "	1193/1588 U1.1
State " GREEN_BLK_ON "	395/1588 U1.13
Substate exited when parent exits	50%
State " GREEN_BLK_OFF "	6/6 U1.1
State " GREEN_BLK_ON "	0/6 

Full Coverage

Model Object	Metric
Transition "[b1_pressed & (p1 == OFF)]" from " GREEN_BLK_OFF " to " GREEN_BLK_ON "	Decision
Transition "[after(GREENDUR, sec).](p1 == ON)]" from " GREEN_BLK_ON " to " GREEN_BLK_OFF "	Decision
Transition "[b1_pressed & (p1 == OFF)]" from " GREEN_BLK_ON " to " GREEN_BLK_ON "	Decision

12. State "[WORKING_TIME](#)"

[Justify or Exclude](#)

Parent: [Gate_Harness/SYSTEM CORE.SYSTEM](#)

Metric	Coverage (this object)	Coverage (inc. descendants)
Cyclomatic Complexity	1	5
Decision	100% (2/2) decision outcomes	100% (6/6) decision outcomes

Decisions analyzed

Substate executed	100%
State "WORKING_TIME_INC"	371/7248 U1.5
State "WORKING_TIME_RESET"	6877/7248 U1.1

Full Coverage

Model Object

Transition "[\[b3_pressed & p2 == OFF\]](#)" from
"[WORKING_TIME_INC](#)" to
"[WORKING_TIME_INC](#)"

Transition "[\[b3_pressed & p2 == OFF\]](#)" from
"[WORKING_TIME_RESET](#)" to
"[WORKING_TIME_INC](#)"

Metric

Decision

Decision